17/01/24 A/5

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[23BSPH011]

FIRST SEMESTER B. Tech EXAMINATION, JANUARY - 2024

ENGINEERING PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Missing data may be suitably assumed.
- **ii.** Physical constants
  - Mass of the electron $m_e = 9.1 \times 10^{-31} kg$
  - Boltzmann constant $k = 1.38 \times 10^{-23} J/K$
  - Charge of the electron $e = 1.602 \times 10^{-19} C$
  - Planck's constant $h = 6.626 \times 10^{-34} Js$
  - Velocity of light $c = 3 \times 10^{8} m/s$
- **iii.** Answer all questions

		PART - A			05 X 02=10
1.	**a**	What are matter waves?	L-2	CO1	**02**
	b.	What is the role of the fully silvered and partially silvered mirror in a resonant cavity?	L-3	CO2	**02**
	c.	Explain Fermi Energy	L-3	CO3	**02**
	d.	Outline the properties of Shock Waves.	L-2	CO4	**02**
	e.	Define Young's modulus and Rigidity modulus.	L-1	CO5	**02**
		PART - B			06 X 05=30
2.	**a.**	State de-Broglie`s Hypothesis. Derive an expression for de-Broglie wavelength.	L-2	CO1	**05**
		OR			
	b.	Using Heisenberg's Uncertainty principle, prove that the electron does not exist inside the nucleus.	L-2	CO1	**05**
3.	**a.**	Explain the physical significance of wave function.	L-2	CO1	**05**
		OR			
	b.	With neat labelled diagrams, explain the interaction of radiation with matter.	L-3	CO2	**05**
4.	**a.**	Explain the construction of an optical fiber with a neat diagram.	L-3	CO2	**05**
		OR			
	b.	Describe the principle of optical fibers.	L-3	CO2	**05**
5.	**a.**	Mention the assumptions of classical free electron theory of metals	L-3	CO3	**05**
		OR			
	b.	Explain the success of quantum free electron theory.	L-3	CO3	**05**

6.	**a.**	Derive the differential equation for Simple Harmonic Motion.	L-2	CO4	**05**
		OR			
	b.	With a neat labelled diagram, discuss how the photodiode conducts in reverse bias.	L-3	CO4	**05**
7.	**a.**	Define stress and strain. Paraphrase the types of elastic moduli.	L-2	CO6	**05**
		OR			
	b.	What is Bending moment of a beam. Mention its applications.	L-4	CO6	**05**
		PART - C		**06 X 10=60**	
8.	**a.**	i. Describe Davisson and Germer`s experiment for confirmation of de-Broglie's hypothesis. ii. Calculate the de-Broglie of a 0.3 kg cricket ball with a speed of 120 km/hr.	L-2 L-3	CO1 CO1	**08** **02**
		OR			
	b.	Show that the de-Broglie wavelength of an electron accelerated by a potential difference V volt is $\lambda = 1.226/\sqrt{V}$ nm. In a measurement of position and momentum that involved an uncertainty of 0.03%, the speed of an electron was found to be 800 m/s. Calculate the corresponding uncertainty that arises in the measurement of its position.	L-4 L-3	CO1 CO1	**06** **04**
9.	**a.**	What are the properties of matter waves? Set up time independent one-dimensional Schrodinger's wave equation.	L-1 L-3	CO2 CO1	**04** **06**
		OR			
	b.	Using one dimensional time independent Schrodinger's wave equation, discuss the solutions for energy of a particle in one dimensional potential well of infinite height.	L-4	CO1	**10**
10.	**a.**	Discuss the three types of optical fiber with neat diagrams.	L-3	CO2	**10**
		OR			
	b.	Derive an expression for numerical aperture in terms of refractive indices of core and Cladding and then arrive at the condition for propagation.	L-3	CO2	**10**

11.	**a.**	**i.**	Using the expression for charge career concentration for electrons and holes, show that Fermi level lies exactly in between the valence band and conduction band in case of intrinsic semiconductor.	L-2	CO4	**06**
		ii.	Use the following given data for intrinsic germanium at 330 K. find the resistivity. $n_i = 2.4 \times 10^{19}/m^3$, $\mu_e = 0.39\ m^2v^{-1}s^{-1}$, $\mu_h = 0.19\ m^2v^{-1}s^{-1}$.	L-3	CO4	**04**
			OR			
	b.		Discuss the procedure to find the band gap of a semiconductor experimentally with necessary equations.	L-2	CO4	**10**
12.	**a.**		Obtain an expression for force constant in case of springs when connected in series and parallel combination.	L-2	CO4	**10**
			OR			
	b.		Give the theory of damped oscillations.	L-2	CO4	**10**
13.	**a.**		Obtain the relation between Young's modulus, rigidity modulus and Poisson's ratio when the force is applied on one face of a cube.	L-3	CO6	**10**
			OR			
	b.	**i.**	Obtain an equation for the Young's modulus in terms of depression when a downward force is applied to beam of a single cantilever. (Use the relation for the bending moment $= \frac{Y}{R} I_g$)	L-3	CO6	**07**
		ii.	When a tangential force of 100 N is applied on one of the faces of cube, it displaces by 2 mm. If the side of a cube is 10 cm, calculate the rigidity modulus of a cube.	L-3	CO6	**03**